THIRD WEEK OF DEVELOPMENT



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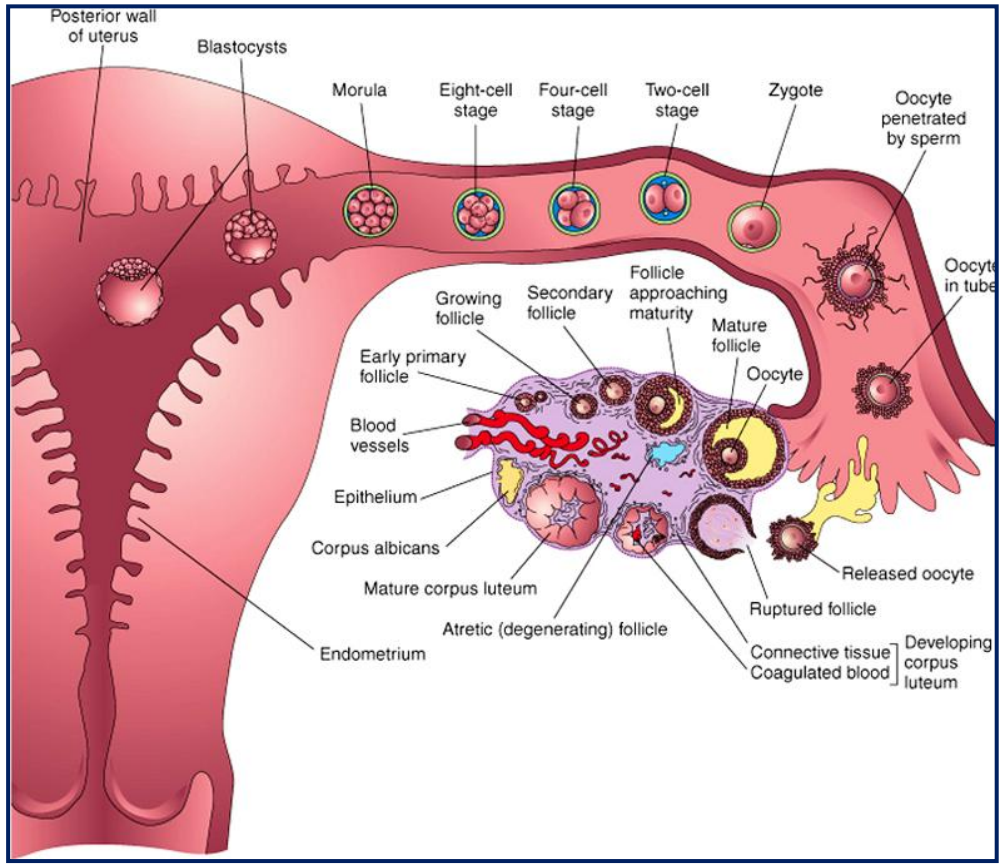
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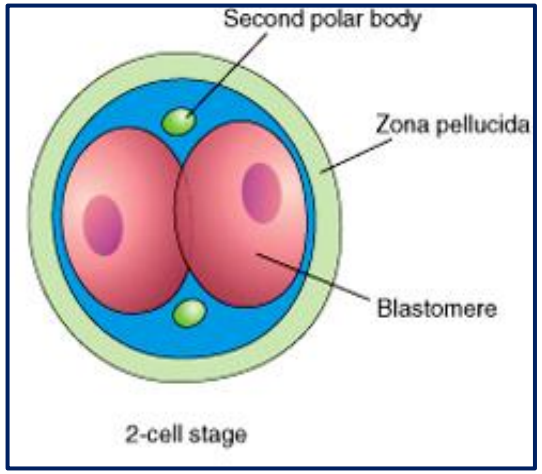
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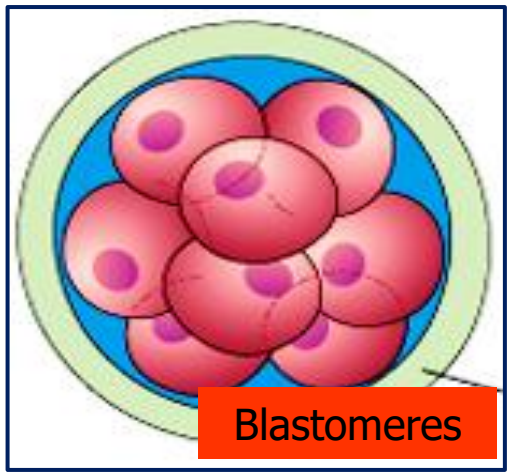
- Describe the process of gastrulation and
- Describe the formation of trilaminar germ disc
- Describe the formation of notochord
- Describe formation of congenital anomalies during 3rd week of development

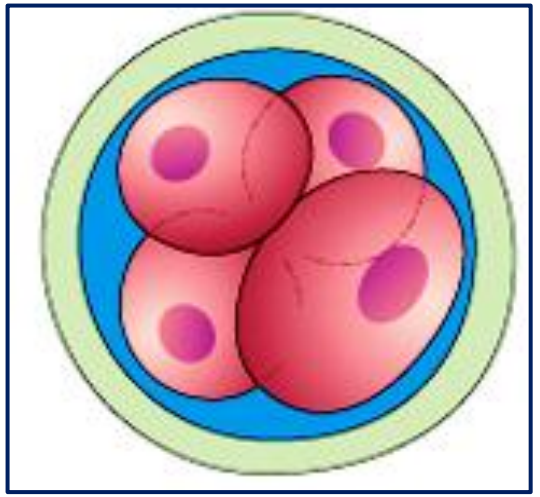
RECAP OF FIRST WEEK OF PREGNANCY

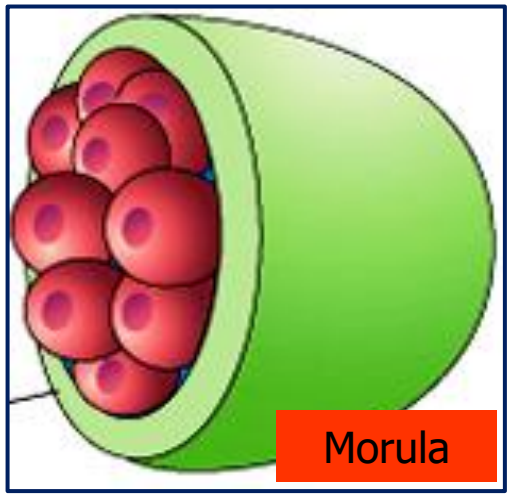
CLEAVAGE







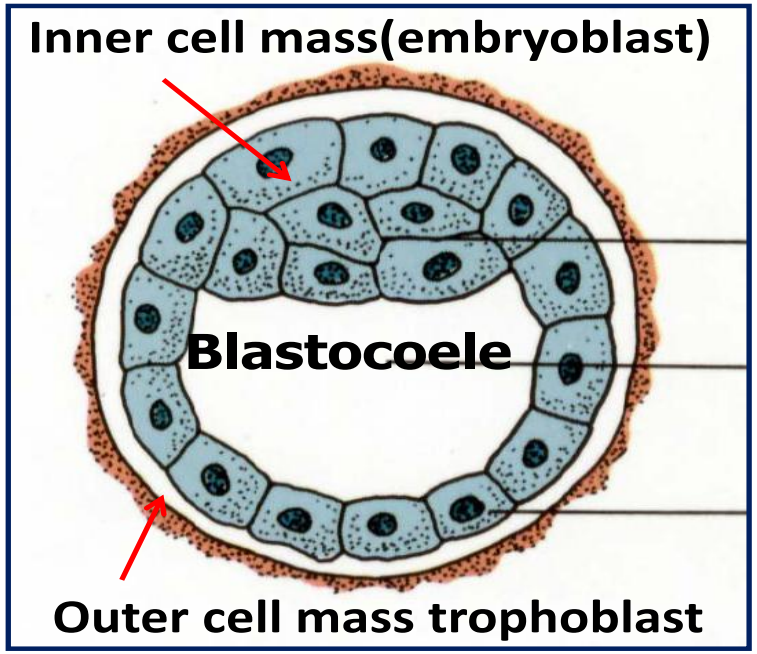


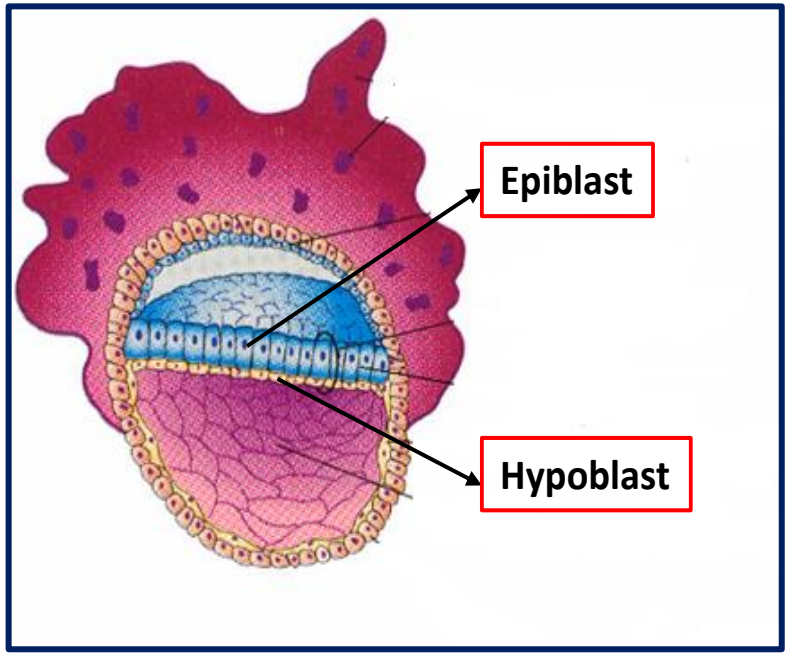


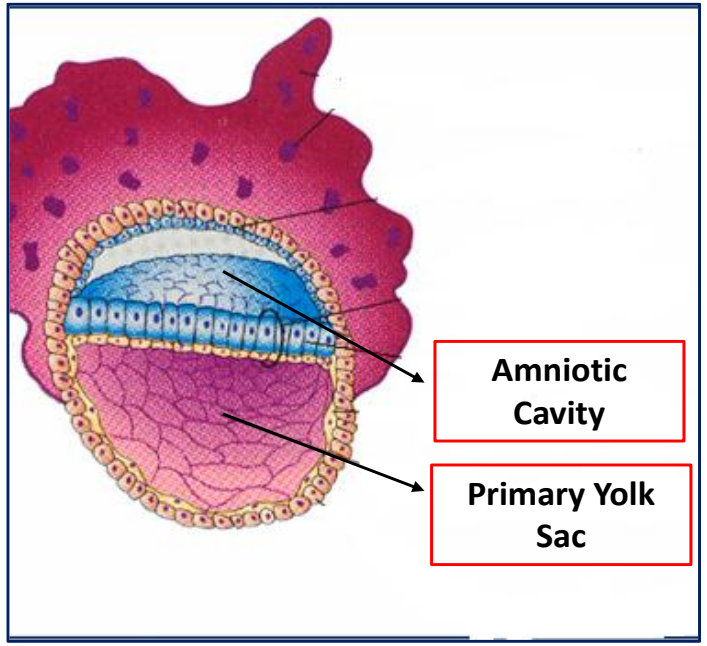
RECAP OF SECOND WEEK OF PREGNANCY

- Morula – 16 cell stage
- Inner cell mass – embryoblast
- Outer cell mass - trophoblast
- Blastocyst

- Implantation ends around 11-12 days after fertilization
- Embryoblast divides into epiblast and hypoblast
- Cavity within epiblast – amniotic cavity
- Cavity within hypoblast - primitive yolk sac

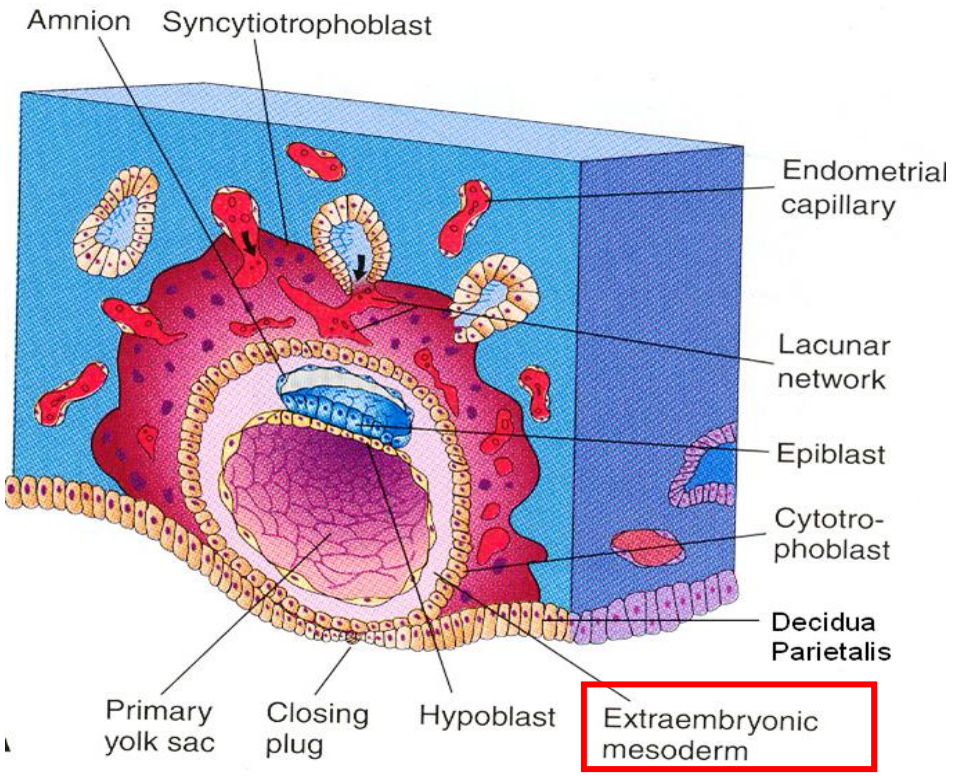




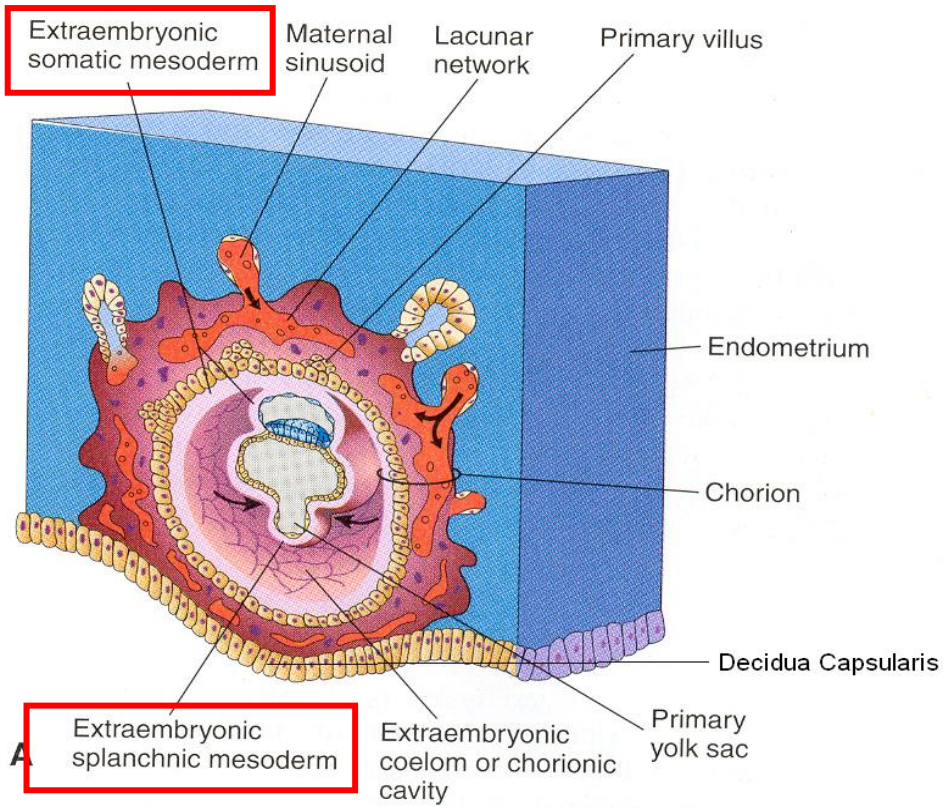


RECAP OF SECOND WEEK OF PREGNANCY

- **Extraembryonic mesoderm (EEM)– cavity within extraembryonic coelom**

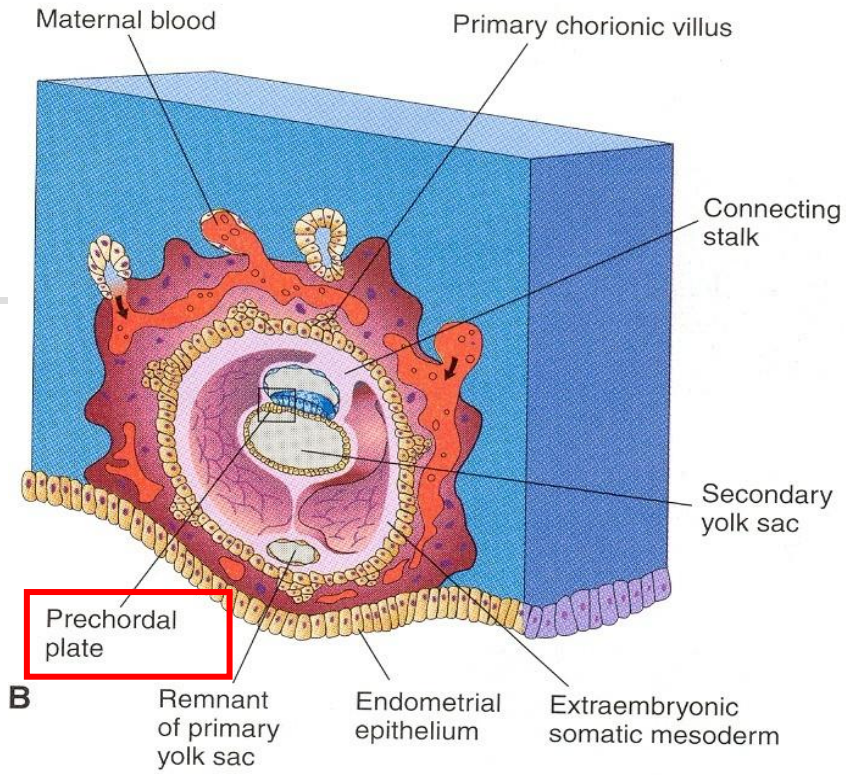


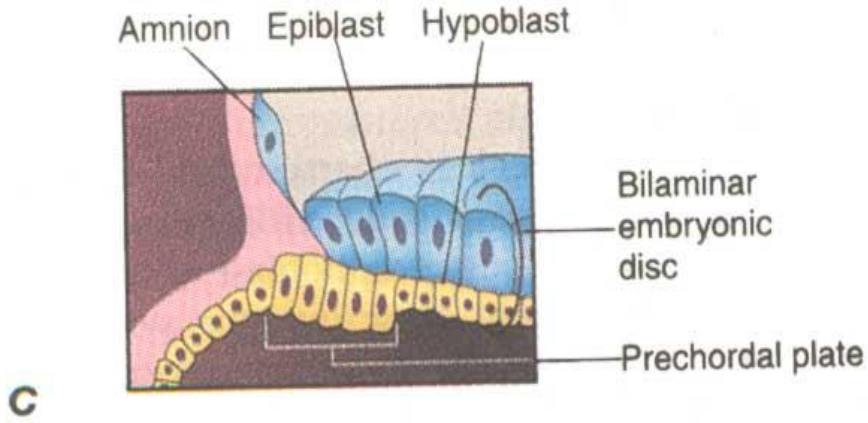
- **EEM divides into somatopleuric and splanchnopleuric mesoderm**



PRECHORDAL PLATE

- At 14th day after fertilization the embryo is still in the form of bilaminar germ disc. Head end, tail end or sides could not be made out.
- At this stage the hypoblast cells become columnar from cuboidal at a localized area.
- This thickened circular area is called – PRECHORDAL PLATE
- Prechordal plate indicates the future head end of the embryo.
- It determines the central axis of the embryo.

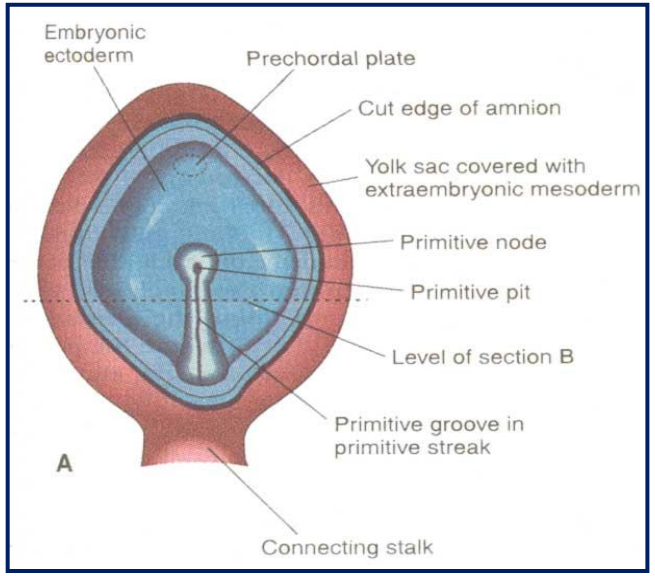


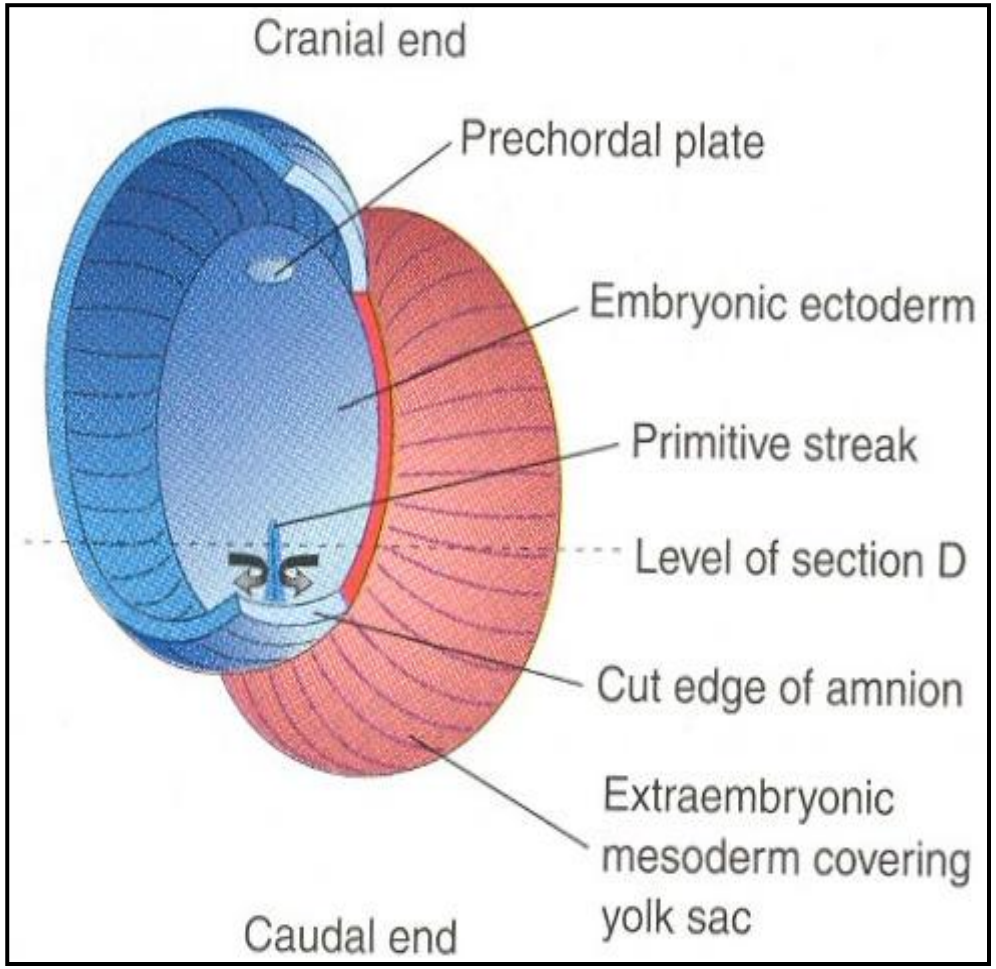


PRIMITIVE STREAK

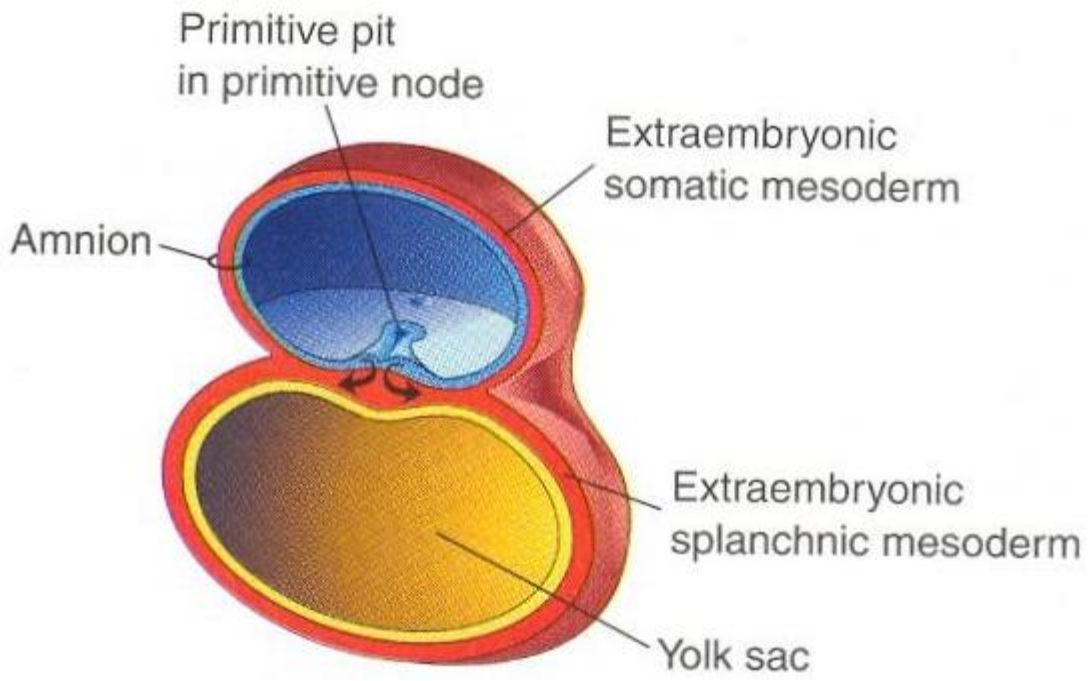
- At the tail end of the embryo the epiblastic cells proliferate & migrate towards the median plane.
- Forms a thickening at the caudal end – PRIMITIVE STREAK.

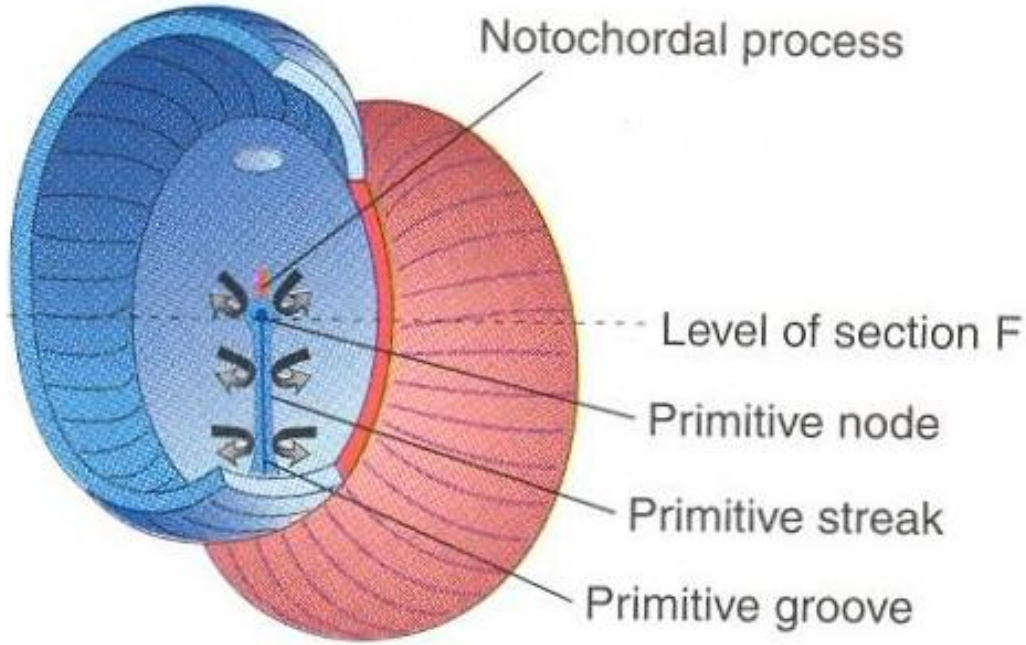
- The cranial end cells proliferate to form a globular swelling – PRIMITIVE NODE OR KNOT.



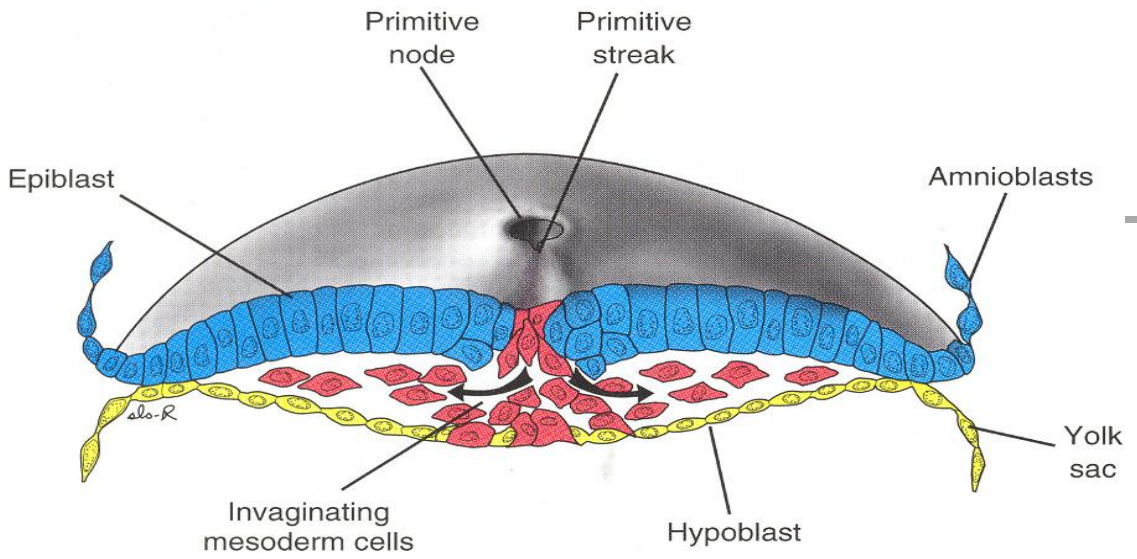


- **A depression is formed in the primitive node – PRIMITIVE PIT OR BLASTOPORE.**
- **It extends into the primitive streak – PRIMITIVE GROOVE.**
- **It determines the central axis, dorsal & ventral surfaces of the embryonic disc.**



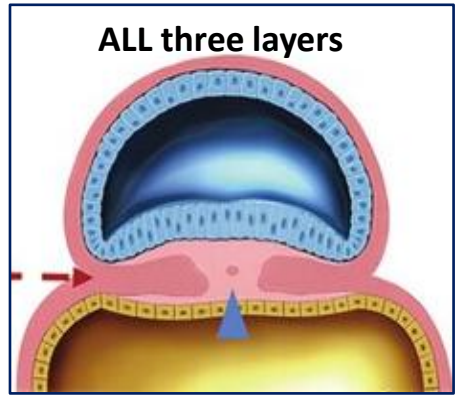


GASTRULATION

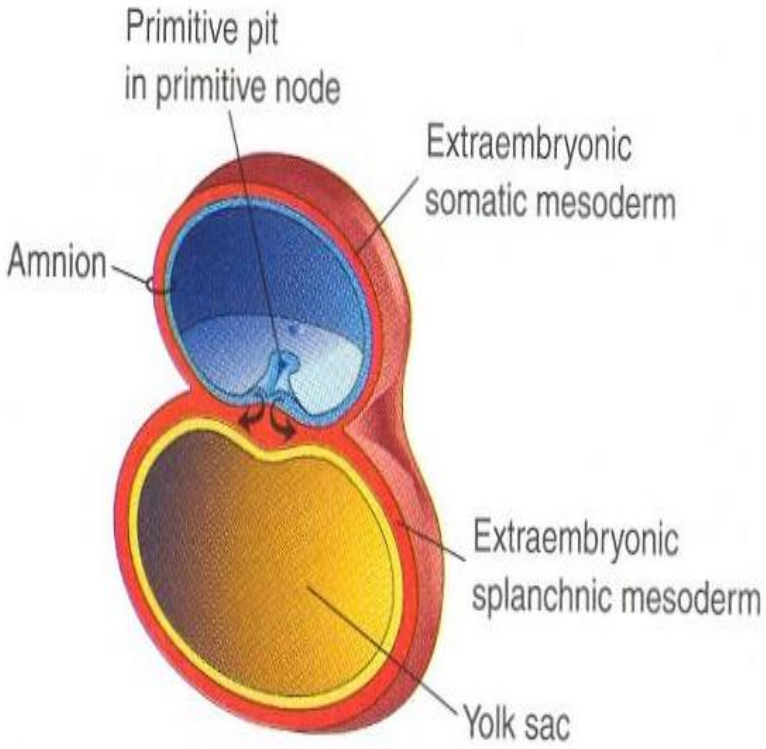


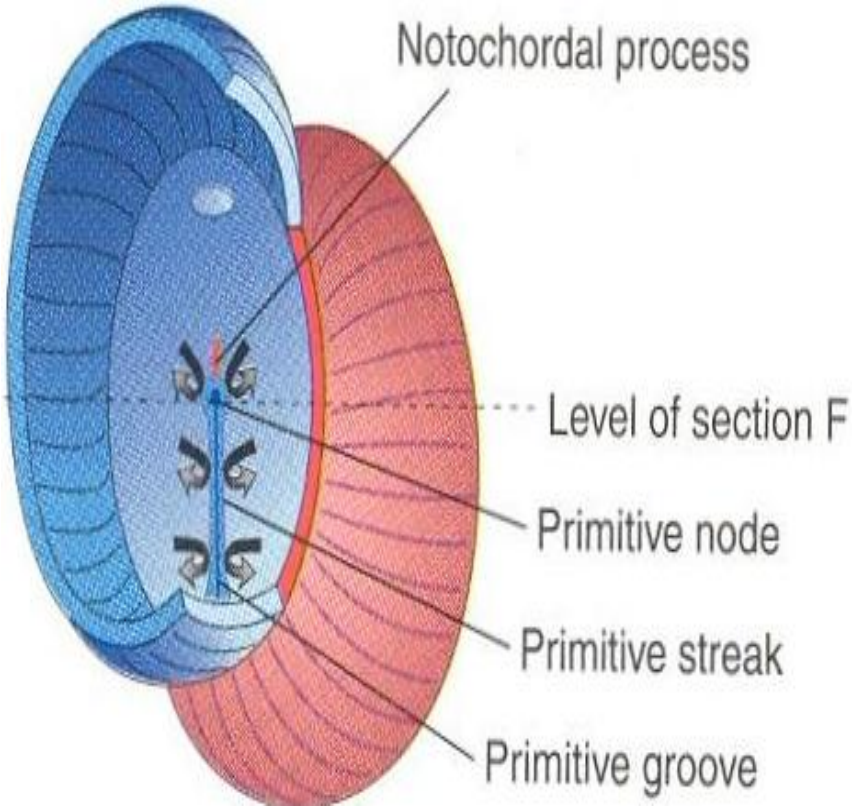
- The proliferating epiblastic cells upon reaching the primitive groove become flask shaped.
- They now invaginate through the groove & spread between epiblast & hypoblast.
- Some of them will replace the hypoblast & forms EMBRYONIC ENDODERM.

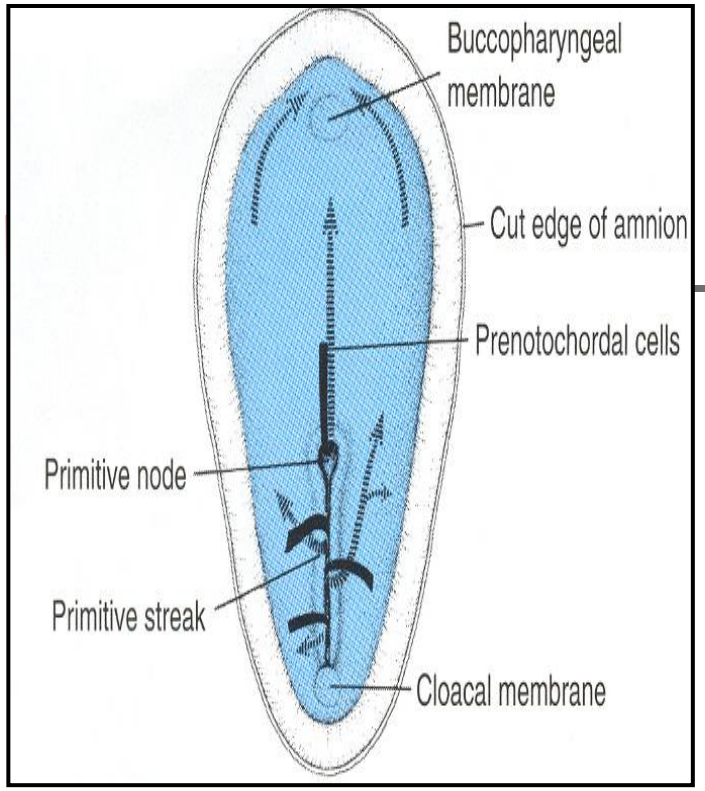
- Others spread between epiblast & newly formed endoderm to form INTRA EMBRYONIC MESODERM.
- The remaining cells of the epiblast will persist as EMBRYONIC ECTODERM.
- So all three germ layers are derived from epiblast ultimately.
- This process of invagination of epiblast cells through the primitive streak & formation of three germ layers is called – GASTRULATION.



- The mesodermal cells spread between the endoderm below & ectoderm above in all directions.
- Gradually they migrate beyond the margin of the disc & merge with splanchnopleuric extra embryonic mesoderm.







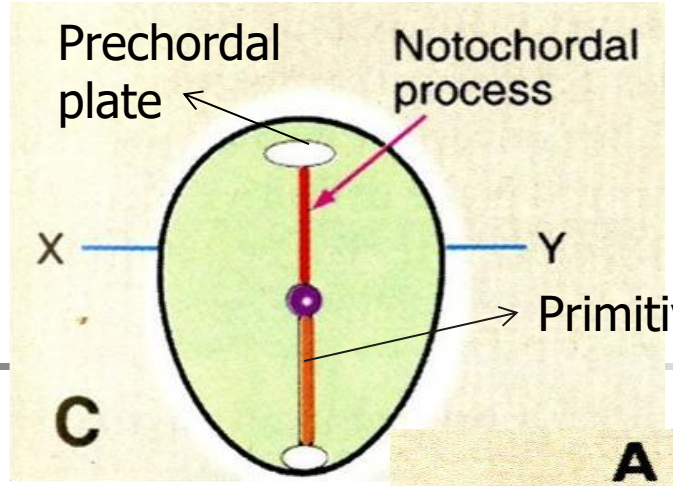
- However in the region of prochordal plate it does not spread.
- Here the ectoderm & endoderm are still in contact with each other – BUCCOPHARYNGEAL MEMBRANE .
- This later degenerates & forms the future oral cavity opening.
- The Mesoderm spreads cranial to prochordal plate & become continuous across the midline- forms the CARDIOGENIC AREA.

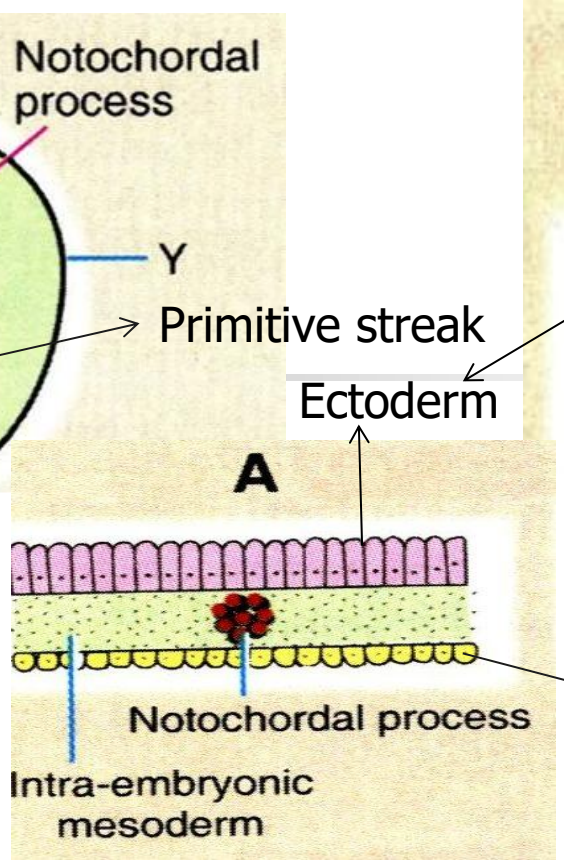
- At the caudal end also behind the primitive streak the mesoderm does not spread between ectoderm & endoderm – CLOACAL MEMBRANE.
- The degeneration of cloacal membrane forms the future anal opening.

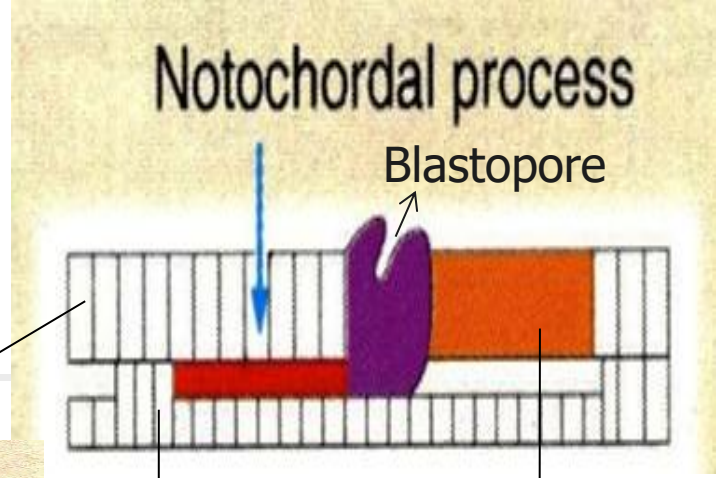
FATE OF PRIMITIVE STREAK

- The primitive streak actively forms mesoderm upto 4th week.
- Thereafter the primitive streak gradually diminishes in size, becomes insignificant & disappears by the end of the fourth week.
- The embryonic disc now containing three germ layers namely ectoderm, mesoderm & endoderm is called – TRILAMINAR GERM DISC.

NOTOCHORD FORMATION



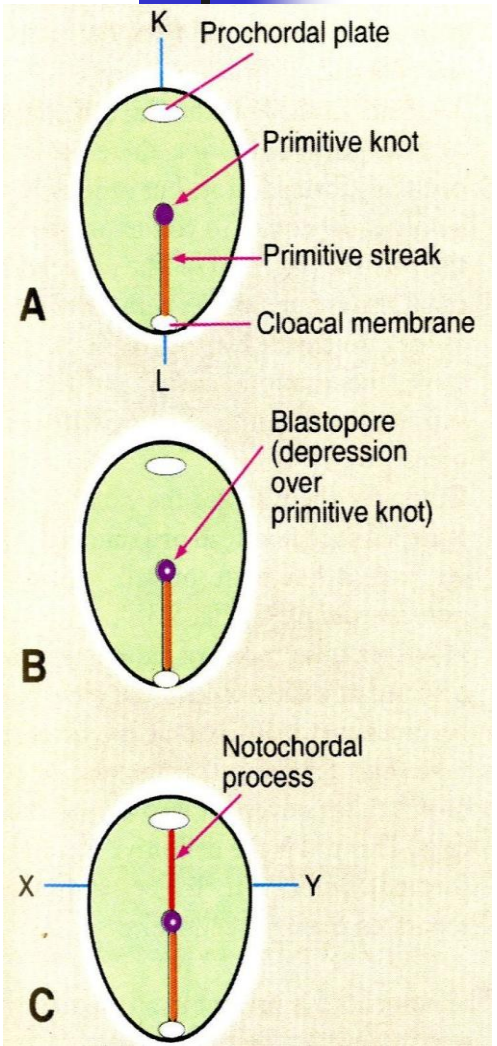




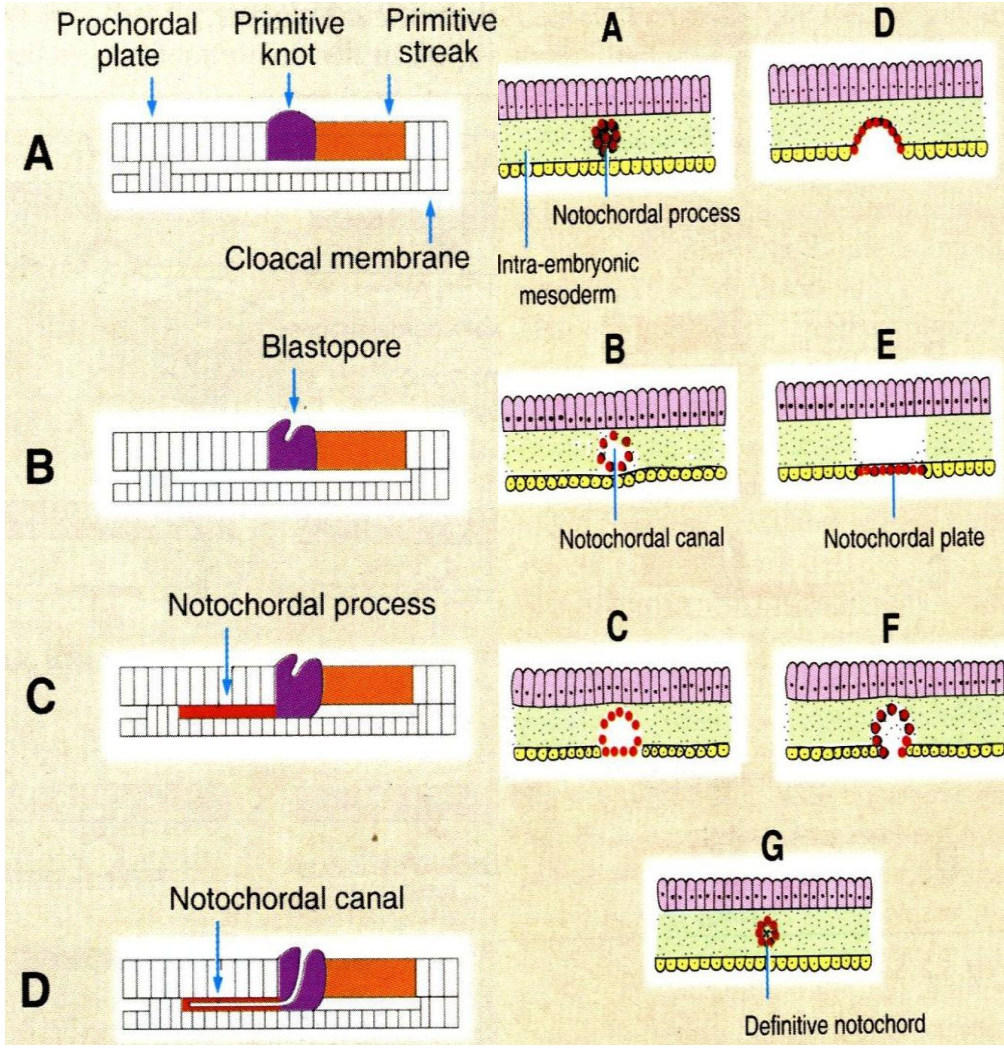
- **Some mesenchymal cells migrate cranially from the primitive node towards the prechordal plate in the median plane.**
- **This solid cords of cells formed in the median plane – NOTOCHORDAL PROCESS.**
- **The cells are called as – PRENOTOCHORDAL CELLS.**

- **This process soon acquires a lumen by the extension of the primitive pit.**
- **Thus it is converted into a tubular structure – NOTOCHORDAL CANAL.**

NOTOCHORD FORMATION

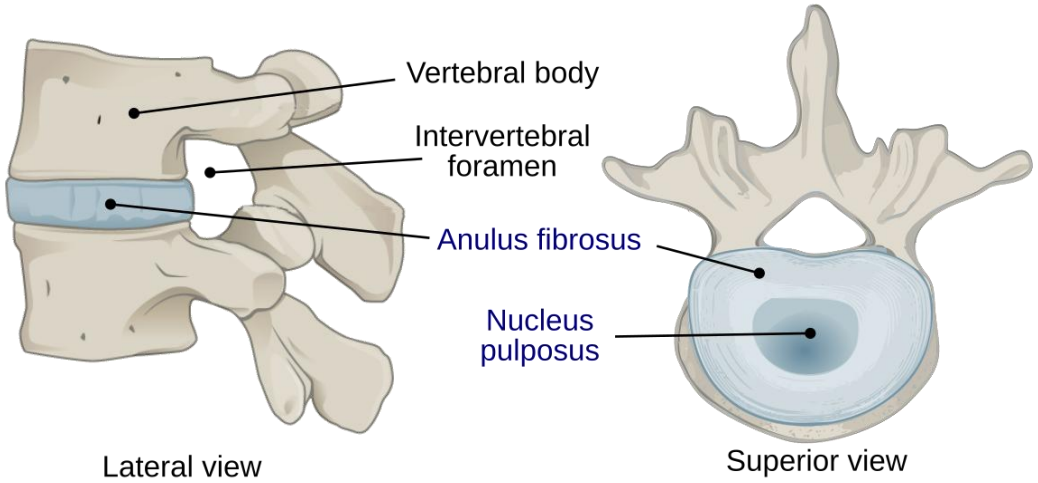


- The cells in the floor of the notochordal canal soon fuses with the endoderm.
- But soon the fused layers degenerate.
- Openings are formed which communicates the notochordal canal with the yolk sac.
- Flatted- Notochordal Plate.
- After sometime the flat notochordal plate soon becomes curved & tubular.
- The notochord gets detached from the endoderm – DEFINITIVE NOTOCHORD.



FATE OF NOTOCHORD

- It degenerates & disappears as the vertebral column is formed.
- But its remnants persist in the intervertebral discs as **NUCLEUS PULPOSUS**.



FUNCTIONAL SIGNIFICANCE OF NOTOCHORD

1. Defines the primordial axis of the embryo.
2. Gives some rigidity to the embryo.
3. Serves as the basis for the development of axial skeleton.
4. Indicates future site of vertebral column.
5. Acts as the primary inductor in the formation of neural tube which forms the future nervous system

Derivatives of Endoderm

Endoderm
(lining of primitive gut tube and its buds)

Digestive System
(Foregut, Midgut, Hindgut)

- **Pharynx (inferior part)**
- **Oesophagus**
- **Stomach**
- **Small intestine**
  - Duodenum (proximal part)
  - Jejunum
  - Ileum
- **Large intestine**
  - Caecum
  - Appendix
  - Ascending colon
  - Transverse colon
  - Descending colon
  - Sigmoid colon
  - Upper part of rectum
- **Anal canal** (epithelium) (below pectinate line)

Respiratory System

- **Larynx** (except the epiglottis)
- **Trachea**
- **Bronchi**
- **Bronchioles**
- **Alveoli of lungs**

Accessory Digestive Organs

- **Liver**
- **Gallbladder and biliary apparatus**
- **Pancreas** (except the dorsal pancreatic bud gives rise to the accessory duct)

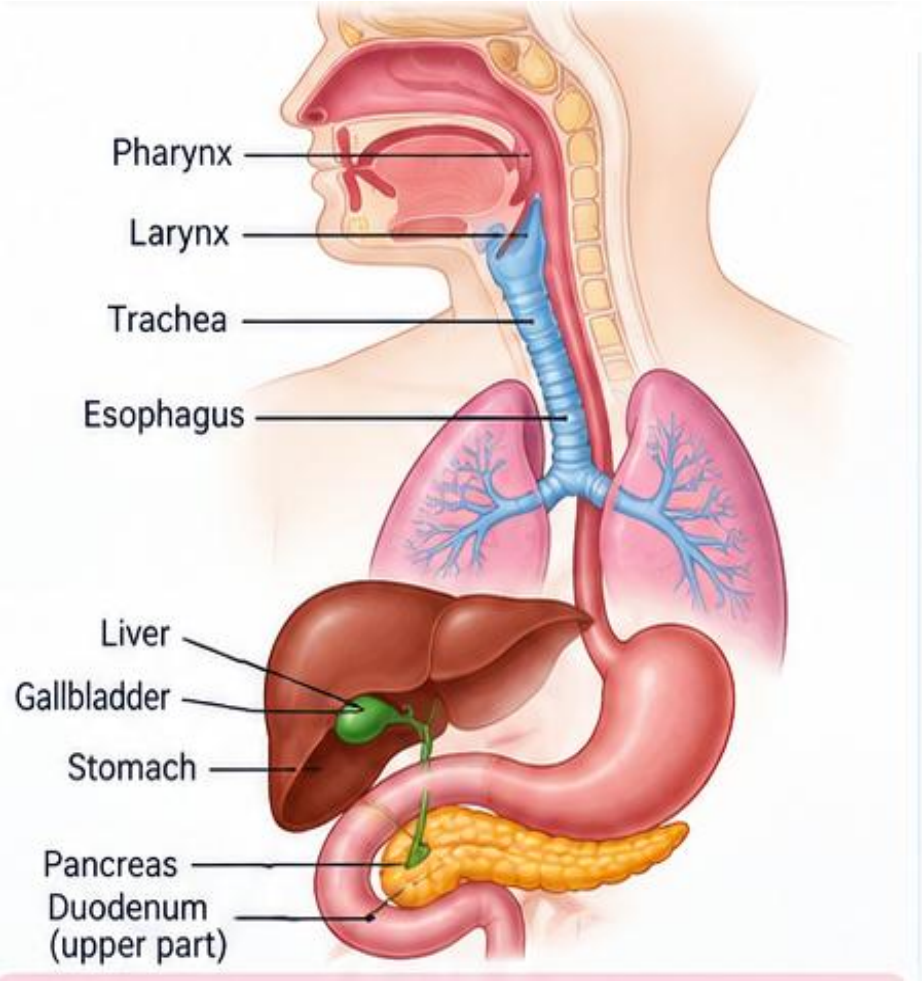
Endocrine Glands

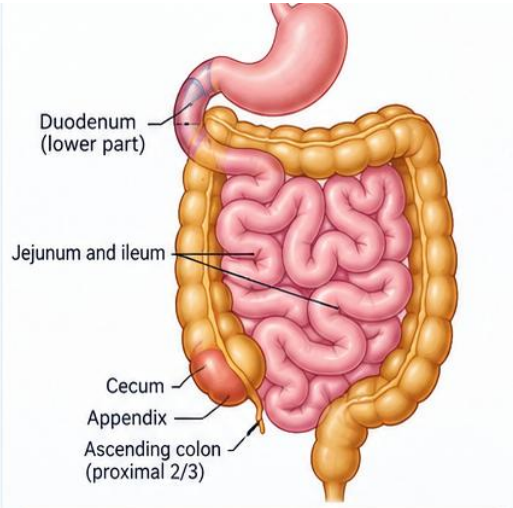
- **Thyroid gland** (follicular cells)
- **Parathyroid glands** (inferior pair)
- **Thymus**
- **Adrenal cortex** (zona glomerulosa and zona fasciculata)
- **Pancreatic islets** (insulin, glucagon, somatostatin, etc.)

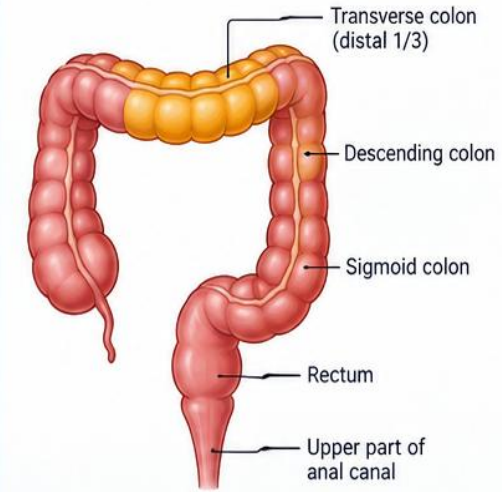
Other Structures

- **Eustachian tube** (auditory tube)
- **Middle ear cavity** and tympanic membrane (mucosa)
- **Tonsils** (crypt epithelium)
- **Urinary bladder** (epithelium)
- **Urethra** (epithelium)

DERIVATIVES OF ENDODERM







UROGENITAL SYSTEM



- Epithelium of urinary bladder (except trigone)



- Most of urethral epithelium
- Prostate glandular epithelium
- Bulbourethral glands epithelium



- Lower part of vagina epithelium



Thyroid gland (follicular cells)



Parathyroid glands (epithelium)



Thymic epithelium



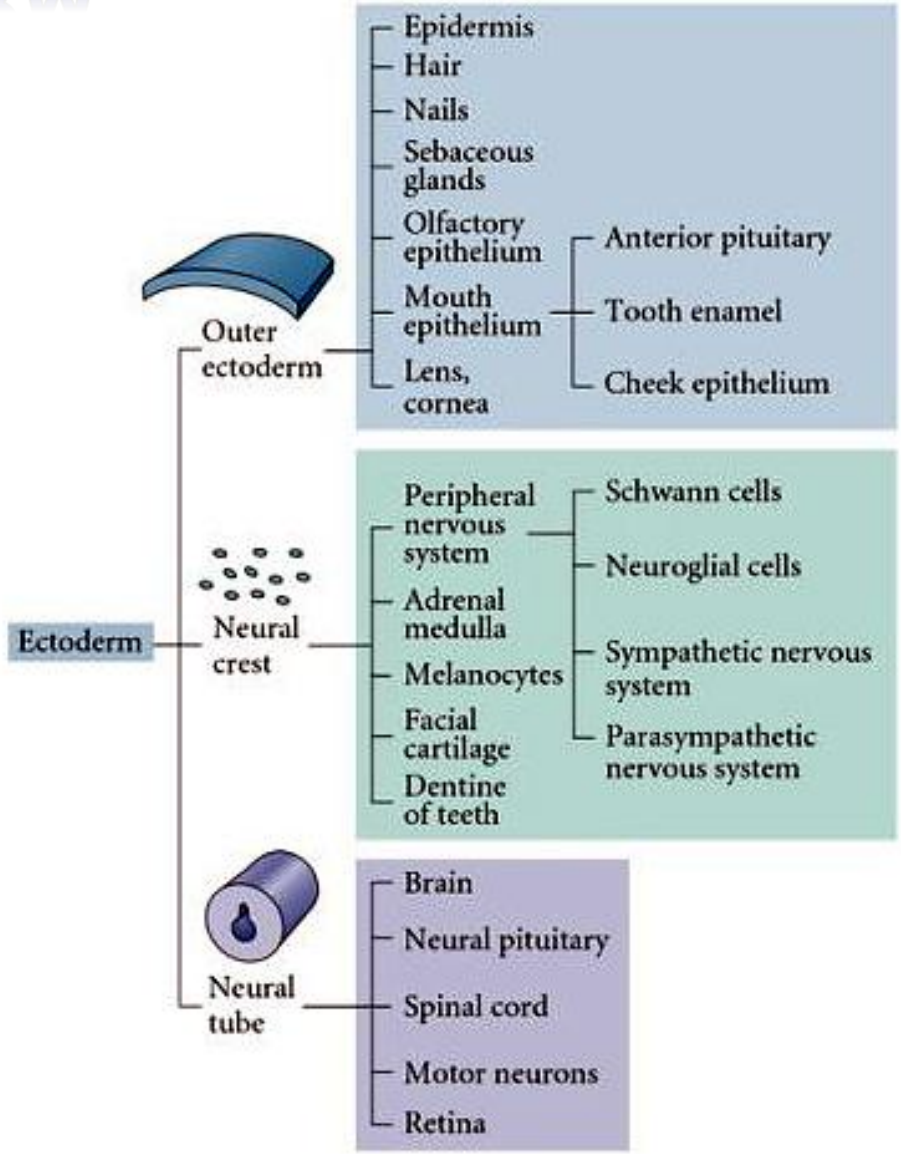
Auditory tube and middle ear epithelium

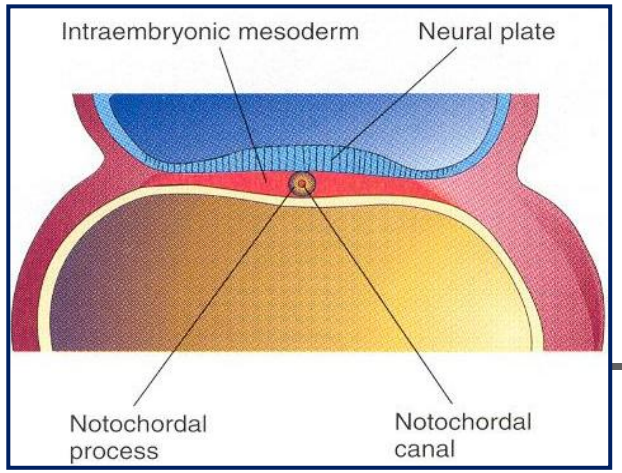


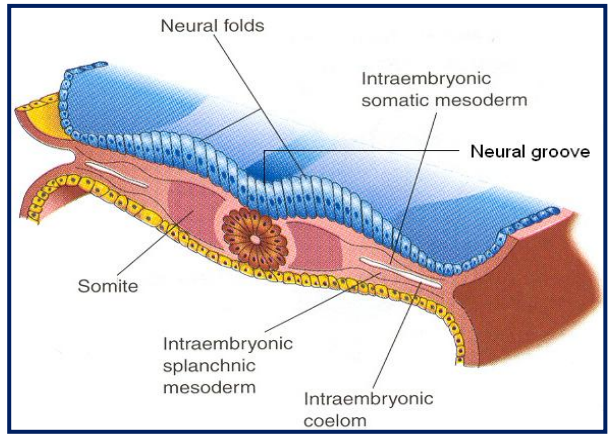
Palatine tonsillar epithelium

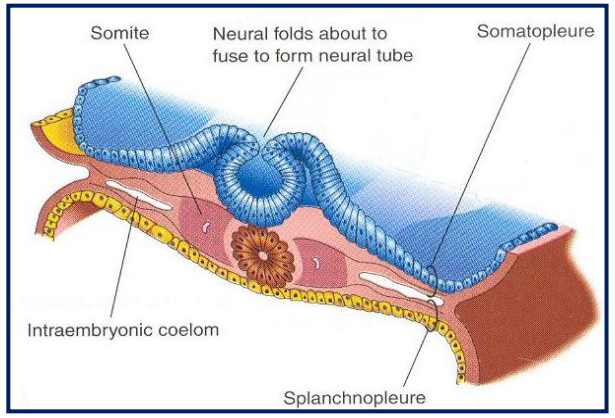
DERIVATIVES OF ECTODERM

1. Epidermis of skin + appendages + mammary glands
2. Nervous system + pituitary gland
3. Special sense receptors in inner ear, nose, and eye
4. Lining of oral and nasal cavities + Salivary glands
5. Lining of lower part of anal canal + terminal urethra
6. Lens + outer surface of cornea + Lacrimal gland
7. Lining of external auditory meatus
8. Derivatives of neural crest (see later)



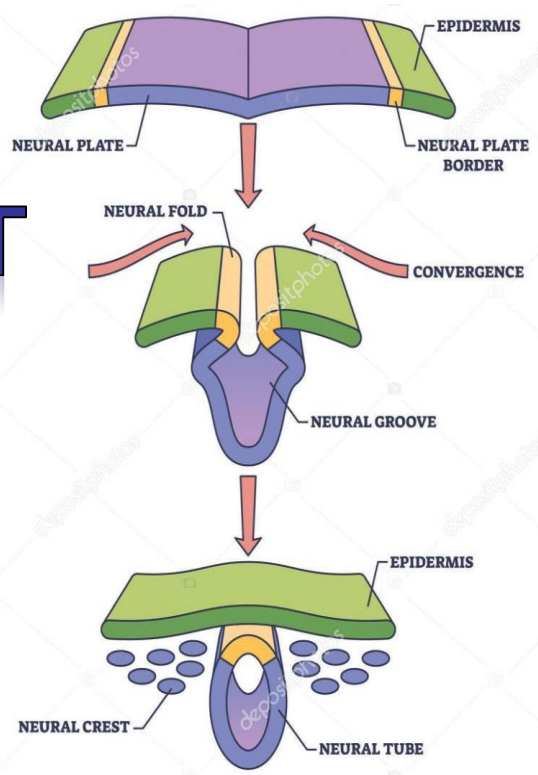


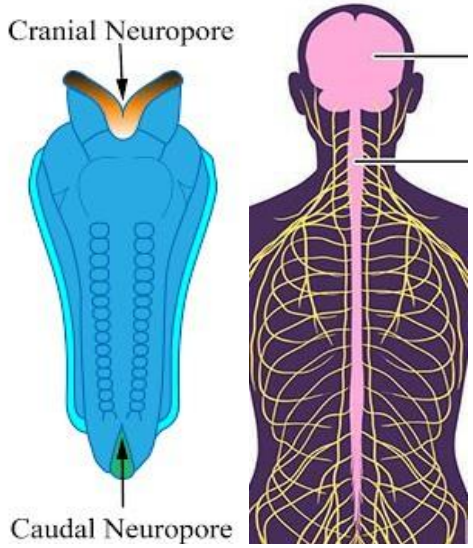




NEURAL TUBE DEVELOPMENT

- **Inducer:** notochord
- **NEURAL PLATE** :the surface ectodermal cells overlying notochord (future brain & Spinal cord).
- **NEURAL GROOVE**.
- The edges of the groove are called as – **NEURAL FOLDS**.
- the neural folds begin to move closer & fuse converting the groove into – **NEURAL TUBE** (proceeds in cranial and caudal directions).
- **Cranial neuropore** closes at day 25 while
- **Caudal neuropore** closes at day 28



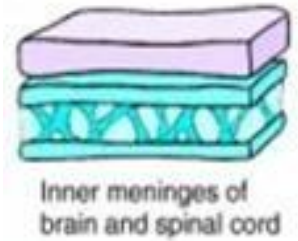


NEURAL CREST CELL

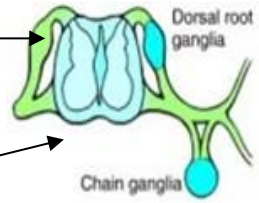
Def: ectodermal cells that separate at the angle between neural folds and surface ectoderm

Derivatives:

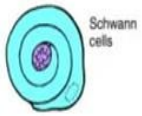
1. Bones of face, laryngeal cartilages (from Pharyngeal arches)
2. Pia and arachnoid (Leptomeninges)



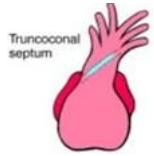
3. Ganglia



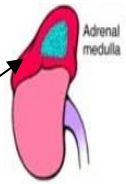
4. Dorsal roots of nerves
5. Schwann cells (for myelination of peripheral nerves)



6. Heart septa

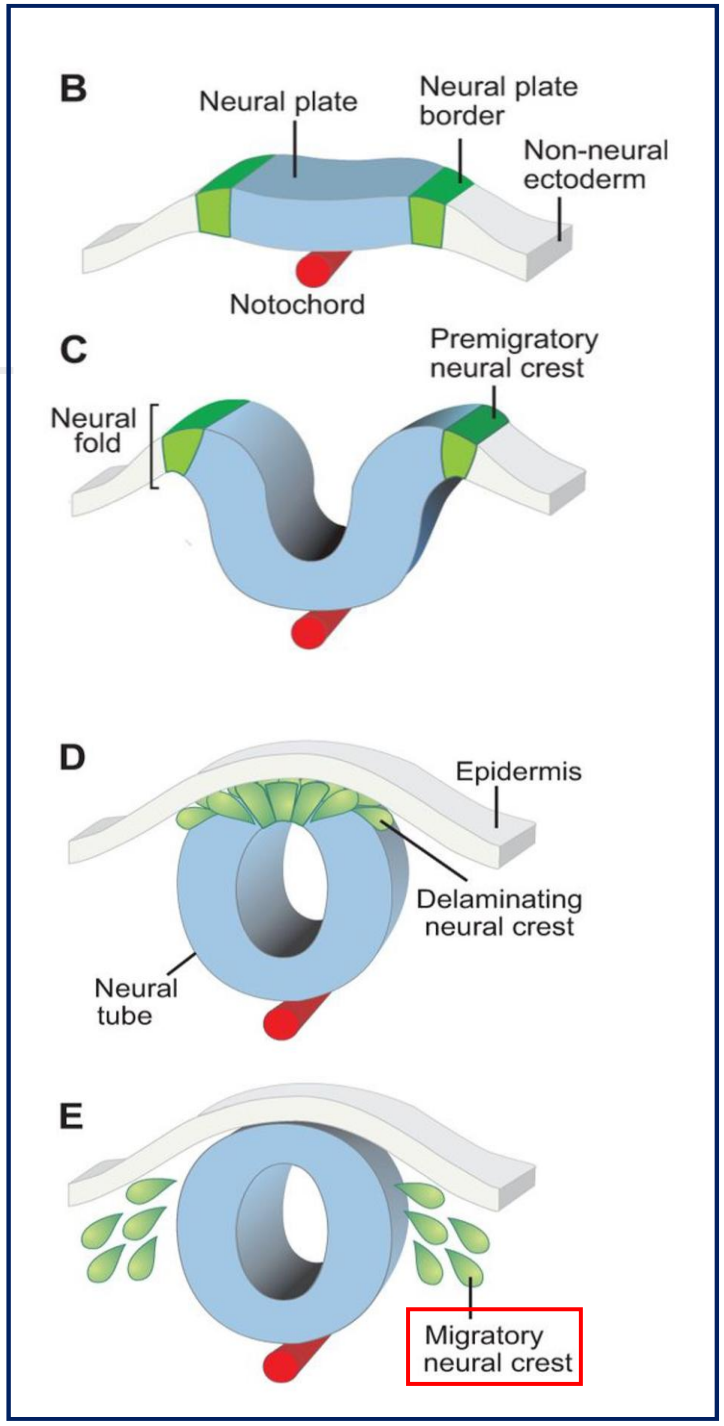


7. Chromaffin cells → Suprarenal medulla



8. Melanocytes





Self study

Teratogenesis Associated with Gastrulation

For example, animal studies indicate that high doses of alcohol at this stage kill cells in the anterior midline of the germ disc, producing a deficiency of the midline in craniofacial structures and resulting in **holoprosencephaly** (see Chapter 17, p. 300). In such a child, the forebrain is small, the two lateral ventricles often merge into a single ventricle, and the eyes are close together (hypotelorism). Because this stage is reached 2 weeks after fertilization, it is approximately 4 weeks from the last menses. Therefore, the woman may not recognize that she is pregnant, having assumed that menstruation is late and will begin shortly. Consequently, she may not take precautions she would normally consider if she knew she was pregnant.

Gastrulation itself may be disrupted by genetic abnormalities and toxic insults. In **caudal dysgenesis (sirenomelia)**, insufficient mesoderm is formed in the caudalmost region of the embryo. Because this mesoderm contributes to formation of the lower limbs, urogenital system (intermediate mesoderm), and lumbosacral vertebrae, abnormalities in these structures ensue.

Affected individuals exhibit a variable range of defects, including hypoplasia and fusion of the lower limbs, vertebral abnormalities, renal agenesis, imperforate anus, and anomalies of the genital organs (Fig. 5.8*A*,*B*). In humans, the condition is associated with maternal diabetes and other causes. In mice, abnormalities of *BRACHYURY (T)*, *WNT*, and *ENGRAILED* genes produce a similar phenotype.

Tumors Associated with Gastrulation

Sometimes, remnants of the primitive streak persist in the sacrococcygeal region. These clusters of pluripotent cells proliferate and form tumors, known as **sacrococcygeal teratomas**, which commonly contain tissues derived from all three germ layers (Fig. 5.9). This is the most common tumor in newborns, occurring with a frequency of 1 in 37,000. Teratomas may also arise from **primordial germ cells** that fail to migrate to the gonadal ridge (see Chapter 1, p. 14).

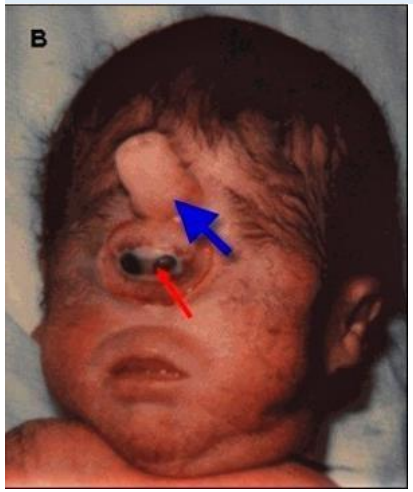




FIGURE 5.8 A,B. Two examples of sirenomelia (caudal dysgenesis). Loss of mesoderm in the lumbosacral region has resulted in fusion of the limb buds and other defects.



FIGURE 5.9 Sacrococcygeal teratoma probably resulting from remnants of the primitive streak. These tumors may become malignant and are most common in female fetuses.

REFERENCES

- Langman's Medical Embryology:
  - Chapter of Third week of development: Trilaminar Germ Disc (Page no 60-72)
  - Third to Eight weeks: The Embryonic Period (page no:74-81)



Thank you!